

Question 1

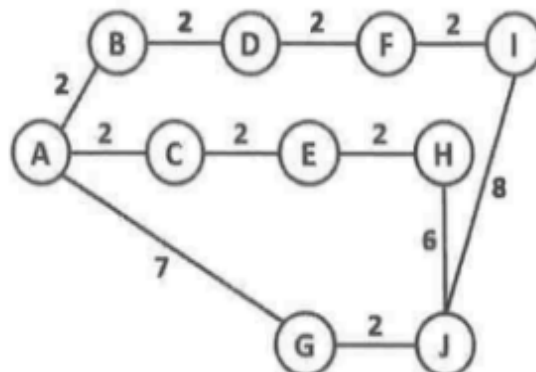


Figure 1-1. Road map of state X

Figure 1-1 shows a road map of 10 different cities represented by nodes A to J in state X, where the label of every arc that connects two nodes represents the distance between them in kilometres, and the straight-line distance between node J and all nodes including node J itself is given in Table 1-1.

Table 1-1: Straight-line distance (in kilometres) between node J and all nodes

A	B	C	D	E	F	G	H	I	J
6	6	5	5	3	4	2	3	4	0

Suppose *Breadth-First Search (BFS)*, *Depth-First Search (DFS)* and *A* Search (A*)* are three search strategies implemented in search of the *shortest path from node A to node J*, where their configurations are provided in Table 1-2, answer the following **THREE (3)** questions:

Table 1-2: Configuration of the three search strategies

Configuration	BFS	DFS	A*
Graph search implementation	✓	✓	✓
Early goal test implementation	✓		
Late goal test implementation		✓	✓
Node Expansion in alphabetical order	✓	✓	✓

- a) Classify *BFS*, *DFS* and *A** as either *uninformed search* or *informed search*. (2 marks)

Uninformed search: BFS and DFS

Informed search: A*

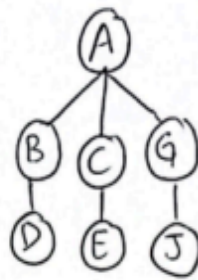
- b) Provide **TWO (2)** similarities between *uninformed search* and *informed search*. (2 marks)

- They return a fixed sequence of actions as a solution.
- They keep track of expanded and unexpanded nodes.
- They explore search spaces systematically.

c) For each of the **THREE (3)** search strategies,

- Construct a *resulting search tree*. (12 marks)
- Identify the *sequence of node expansion*. (3 marks)
- Identify the *solution returned*. (3 marks)
- Compute the *cost of the returned solution*. (3 marks)

BFS:



Traversal Path:

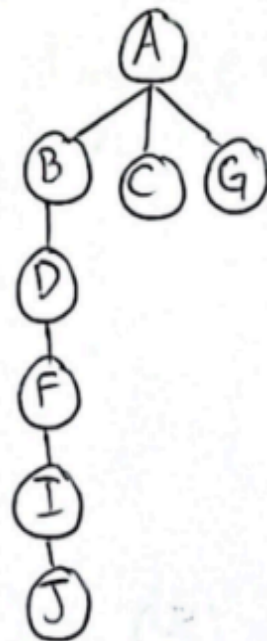
A, B, C, G

Solution Path:

A, G, J

Cost: $7 + 2 = 9$

DFS:

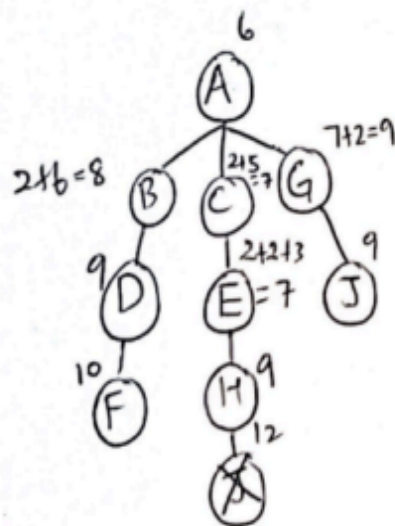


Traversal Path:
A, B, D, F, I, J

Solution Path:
A, B, D, F, I, J

Cost:
 $2+2+2+2+8=16$

A*



Traversal Path:
A, C, E, B, D, G, H, J

Solution Path:
A, G, J

Cost: $7+2=9$

Question 2



Figure 2-1. Two chess agents are playing chess against each other

- a) *Game playing* is a popular application of Artificial Intelligence (A.I.) that involves development of computer programs that can play games such as *chess* (see **Figure 2-1**). These programs process *states* and *actions* from time to time so that they can reason about what to do next and make decisions that will lead to winning outcomes.
- (i) *Agent* is a term that is commonly used to refer to computer programs that are equipped with some kind of intelligence. Explain the meaning of this term in your own words and illustrate your explanation using the diagram provided in **Figure 2-2**. (2 + 3 marks)

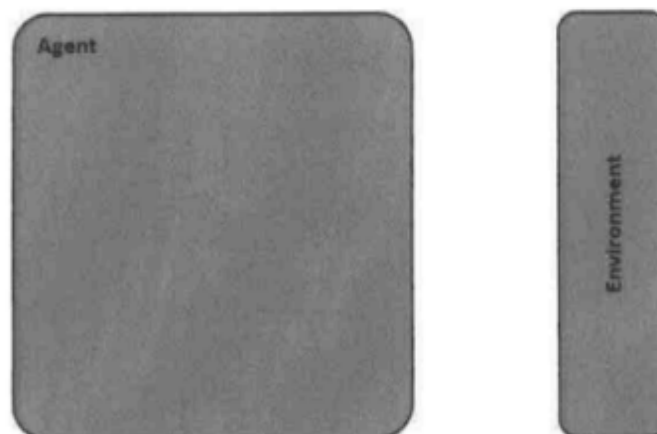
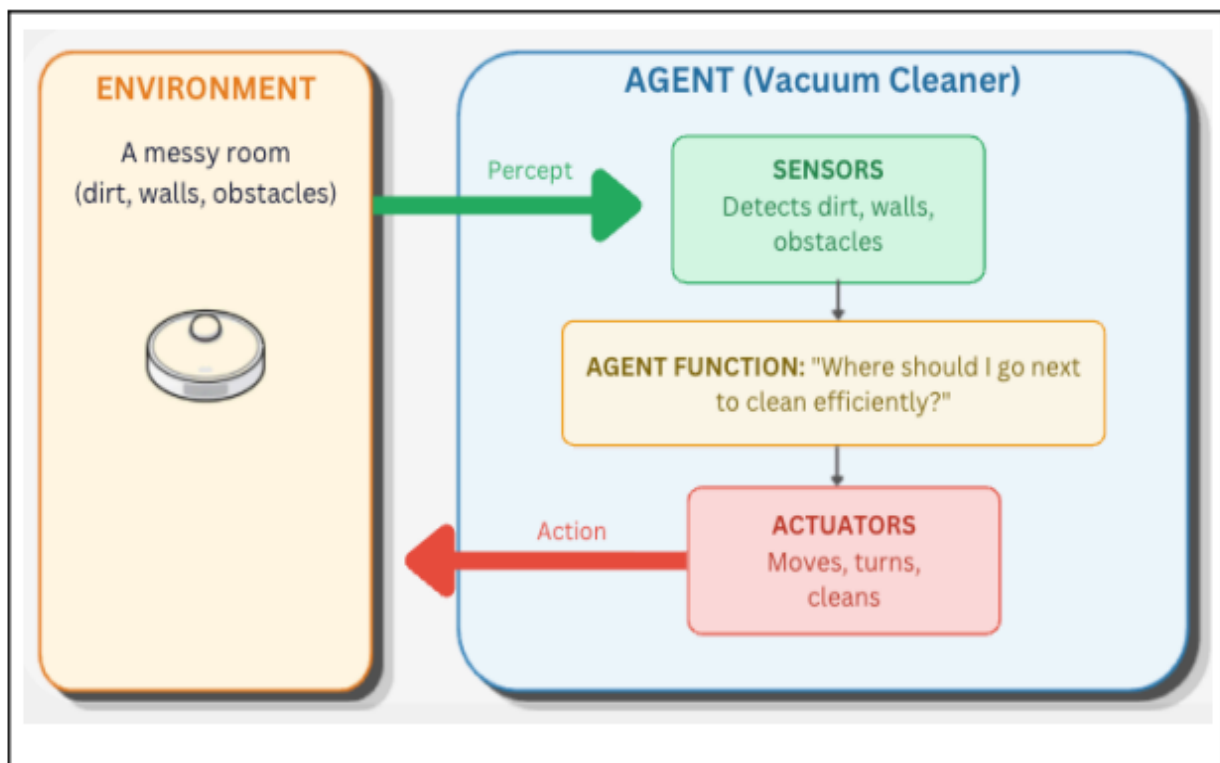


Figure 2-2. Agent and environment

An agent is anything that can be viewed as **perceiving its environment through sensors** and **acting upon that environment through actuators**



- (ii) For most of the games in A.I., a different term is often used to refer to an *action* as well as a *state*. Identify these two terms. (2 marks)

- Action = move
- State = position

- (iii) Games that are most studied in A.I. usually have **FIVE (5)** common characteristics: *two-player*, *turn-taking*, *perfect information*, *deterministic*, and *zero-sum*. Briefly explain the meaning of each of these characteristics. (5 marks)

Turn taking: Only one player moves at a time, then waits for the other's move.

Zero-sum: one's gain = exactly another's loss.

Two-player	A game that is played between two players.
Turn-taking	Players take turns making moves until the game ends.
Perfect information	Both players can see the complete state of the game (Fully Observable).

Deterministic	The same action always produces the same result; there is no randomness.
Zero-sum	One player's gain is the other player's loss. There is no win-win outcome.

Question 3

FingerTec Face Recognition Face ID 4 is a *facial recognition system* (see **Figure 3-1**) designed by a local company to automate daily sign-in procedure of a company. To make this works, a high-quality video camera can be installed at the reception area facing the entrance of an office to scan the face of staff so that the system can recognise and identify their identities as they enter.



Figure 3-1. Facial recognition technology

- a) *PEAS*, stands for *Performance, Environment, Actuator* and *Sensor*, describes how an intelligent agent works. Suppose the lighting of the reception area is turned on 24/7 and adjusted to optimise the performance of the *facial recognition system*, and all users are aware of the need of facing directly into the video camera to increase its chances of success, construct a *PEAS* description for the problem to which the system is going to solve. (4 marks)

Performance measure	Accuracy of recognizing human face
Environment	Images / photographs
Actuators	Display screen (to display the scanned image with recognized faces)
Sensors	Camera

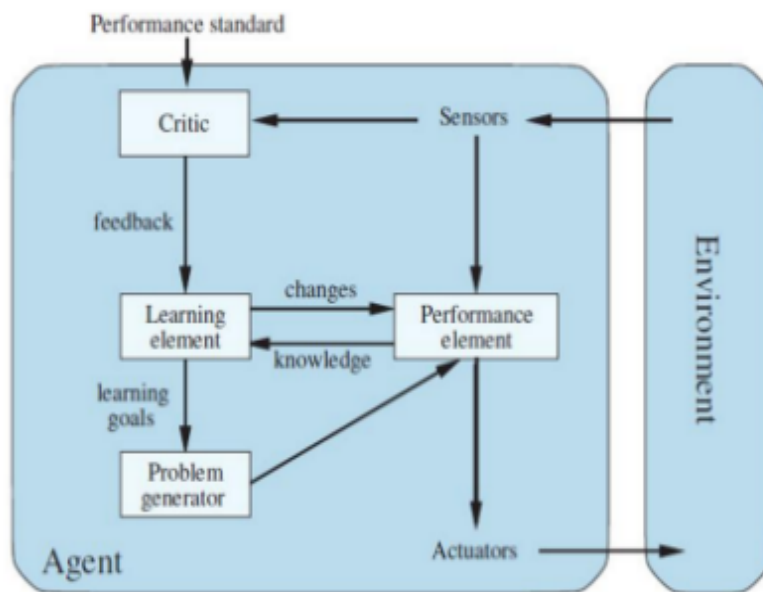
- b) Identify **SIX (6)** properties of the *facial recognition system*'s environment based on information given below and justify your answer. (6 + 6 marks)
- Fully observable, partially observable or non-observable
 - Episodic or sequential
 - Single-agent or multi-agent
 - Static or dynamic
 - Deterministic or non-deterministic
 - Discrete or continuous (in terms of percept)

Fully observable	The agent can make an optimal decision on face recognition because it can "see" the entire image the whole time.
Single agent	The agent is solving face recognition problem by itself.
Deterministic	The outcome of tagging people or faces in an image can always be determined.
Episodic	The result of face recognition for an image has no impact on the result of face recognition for the next image.
Static	When the agent is screening the image for percept or analyzing the image trying to identify faces, the image remains unchanged.
Discrete	Regardless of the resolution of the image be processed, there is a finite set of percepts in terms of pixels.
Unknown	How the agent "sees" an image is different from humans, meaning that the agent cannot really understand what "faces" are like how humans do.

- c) Suppose the *facial recognition system* is a *learning agent* consisting of some learning elements, illustrates how a *learning agent* works using the diagram given in **Figure 3-2**. (9 marks)



Figure 3-2. A learning agent and its environment



Performance element	• It is responsible for taking in percepts and selecting external actions. • It can be considered as the entire non-learning agent. What it does: This part of RoboClean actually does the cleaning. It takes input from its sensors (like bumping into walls or detecting dirt) and decides what to do next (like turn, move forward, or stop). Example: RoboClean sees dirt in front of it → it moves forward and starts cleaning.
Learning element	• It is responsible for making improvements. • It uses feedback from the critic on how the agent is doing to determine how the performance element should be modified to do better in the future.

	What it does: This part helps RoboClean learn from experience. It looks at how RoboClean is doing and adjusts its behavior to clean better next time. Example: RoboClean notices that going under the table always gets stuck → it learns to avoid that area next time.
Critic	• It is responsible for performance evaluation. • It tells the learning element how well an agent is doing with respect to a fixed performance standard. What it does: This part checks how well RoboClean is performing based on a standard (like how much dirt was cleaned or how long it took). Example: After a cleaning session, it gives feedback: “You missed a dirty corner” or “You took too long.”
Problem generator	• It is responsible for identifying any part in need of improvement and suggest exploratory actions that will lead to new and informative experiences. What it does: Suggests trying new things so RoboClean can improve. Example: “Hey RoboClean, try taking a new route around the sofa. Maybe it's faster!”